

A tool for designing efficient guide RNAs in CRISPR-Cas systems

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Group: A4130

Educational program: Chemistry and AI

Type of practice: Research Project

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Glossary

CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) — a bacterial genomic region used in pathogen defense

Guide RNA (gRNA) — the RNA component of the CRISPR-Cas9 genome engineering tool. Contains a ~20 base pair sequence that is complementary to the genomic target.

CRISPR-associated endonuclease protein (Cas) — a family of proteins able to cleave nucleic acids. Cas9 is the most commonly used Cas protein in CRISPR experiments.

Protospacer adjacent motif (PAM) — a short sequence of nucleotides that must be present downstream of the target site for the nuclease to make a double strand break.

GC-content (guanine-cytosine content) is the percentage of nitrogenous bases in a DNA or RNA molecule that are either guanine (G) or cytosine (C).

Indel — the insertion or deletion of nucleotides in the genome.

Off-target effects or off-target activity — Cas9 cleavage at undesired locations due to gRNA targeting sequence with sufficient homology to recruit Cas9 to unintended genomic locations

Target locus — Genomic target of the gRNA. The sequence includes the unique ~20 bp target specified by the gRNA plus the genomic PAM sequence

Addgene, *CRISPR eBook* (3rd ed., 2021), addgene.org [\[Link\]](#)

CRISPR-Cas system

A simple two-component system that can be used to target a specific genomic sequence and then make a cut in the DNA

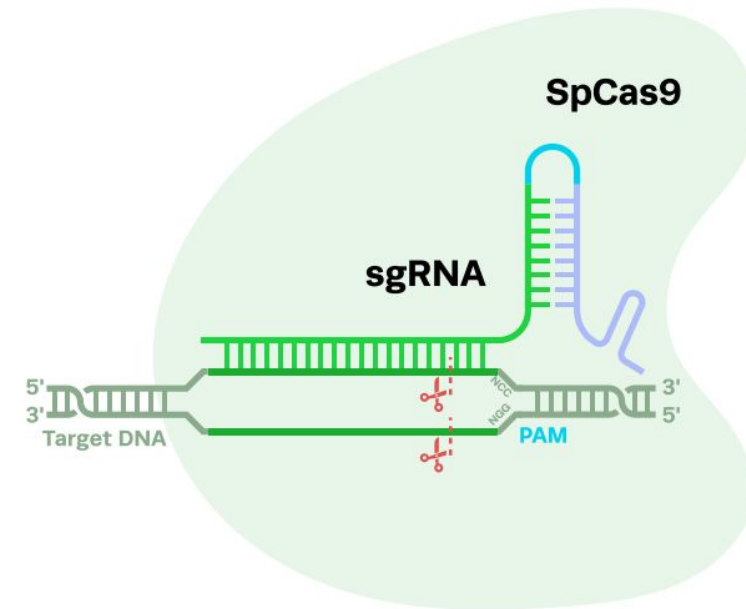
Components:

- a guide RNA (gRNA) that is specific to the target DNA sequence
- a CRISPR-associated endonuclease (Cas)

Protospacer adjacent motif (**PAM**): a short sequence of nucleotides that must be present downstream of the target site

Process:

- gRNA guides the system to a specific DNA site
- Cas protein cuts DNA in the right place



*Minimum required components
of a CRISPR/Cas system*

Relevance

Capabilities

- **Knock-out** – permanent inactivation of a gene via DNA cleavage
- **Knock-in** – integration of a desired DNA sequence into a target gene
- **DNA visualization** using fluorescent proteins
- **Gene regulation** – activation or suppression of gene expression
- **Base editing** – precise replacement of a single nucleotide pair without DNA cleavage

Limitations

- Delivery of CRISPR components into cells
- **Off-target activity**
- **Insufficient editing efficiency**
- Ethical concerns

Application

Research

Studying gene function
Creating mutant cell lines

Biotechnology

Enhancing traits of agricultural animals and plants
Controlling the spread of animal-borne infections
Engineering microbes for biomanufacturing

Therapy

Treating genetic disorders (via ex vivo cell editing)
Potential for cancer therapy

AI for CRISPR

AI is increasingly integrated into all **10 key tasks** of the CRISPR workflow, helping to reduce reliance on costly and time-consuming wet-lab experiments

These tasks are commonly modeled as **binary/multiclass classification, regression**, or even **reinforcement learning** problems, depending on the biological context

AI unlocks faster, cheaper, and more precise guide RNA design and validation

Problem	Count
CRISPR arrays	2
CRISPR loci	2
CRISPR systems CrRNA	2
Acr proteins	6
Aca proteins	1
CRISPR operons	2
Cas protein	2
Off target activity	10
On target activity	16
Editable target region	1
Gene editing outcome prediction	4

Goals & Objectives

Goal

Create an end-to-end software tool that automatically designs, ranks and refines highly efficient, low-off-target guide RNAs for any CRISPR-Cas experiment

Objectives

1. **Candidate mining** – scan target gene(s) for PAM-adjacent 20-nt windows, filter by basic sequence rules (GC%, homopolymers).
2. **Off-target scanner** – genome-wide search (0-4 mismatches) + specificity scoring to discard risky guides.
3. **Efficacy predictor** – develop a hybrid CNN + LSTM model fusing one-hot and RNA-FM embeddings to forecast relative activity.
4. **Sequence refinement loop** – apply reinforcement learning to iteratively mutate guides and maximise (efficacy – off-target) score under hard constraints.

(Optional future) 5. Web/UI & API integration for wet-lab users.

Relative activity predictor

Dataset¹:

- perfectly matched sgRNA (PMsgRNA)
 - mismatched sgRNA (MMsgRNA)
 - Activity of MMsgRNA in comparison to PMsgRNA (relative activity)
- 26248 rows

Representation	Architecture	R2
OHE (2 × 4 × N)	CNN (<i>ExsgRNA</i>)	0.71 (SOTA²)
OHE 7 × N (Zhang et al., 2022)	CNN (<i>ExsgRNA</i>)	0.71
OHE 7 × N + RNA-FM embeddings (N × 640)	CNN (<i>ExsgRNA</i>) + LSTM	0.72
CRISPR-BERT embeddings + OHE 7 × N	CNN + BiGRU (<i>CRISPR-BERT</i>)	0.66
OHE 7 × N	CNN	0.64
RNA-FM embeddings (N × 640)	LSTM	0.43

¹M. Jost et al. (2020) Titrating gene expression using libraries of systematically attenuated CRISPR guide RNAs

²W.-X. Hu et al. (2022) *ExsgRNA*: reduce off-target efficiency by on-target mismatched sgRNA

Relative activity predictor: Final version

Input Features

The sgRNA-DNA pair is represented in two forms:

One-hot encoding — for the CNN branch (from ExsgRNA)

RNA-FM embeddings — for the LSTM branch (inspired by DeepFM-CRISPR)

Architecture Overview

CNN Branch

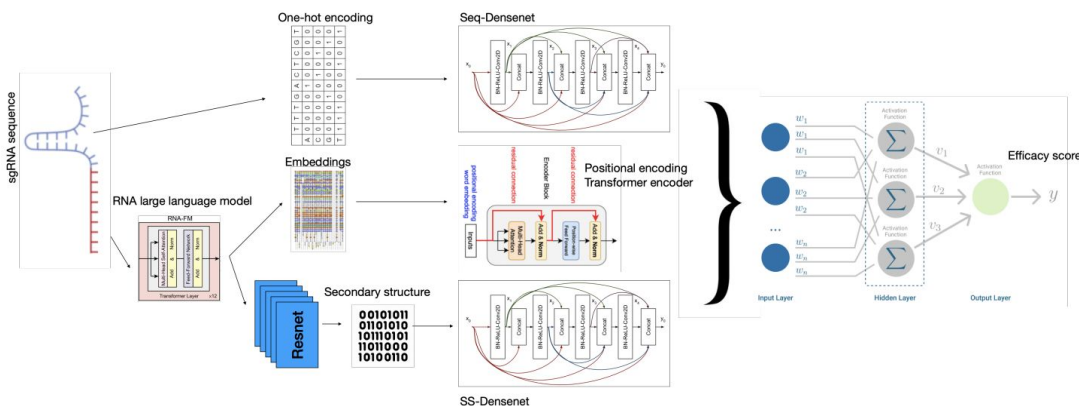
- Input: one-hot ($7 \times 26 \times 1$)
- Two Conv2D layers + MaxPooling → Dense layers
- Captures structural interaction features

LSTM Branch

- Input: RNA-FM embeddings (26×1280)
- LSTM → Dense layers
- Captures contextual sequence semantics

Feature Fusion & Output

- Concatenate CNN and LSTM features
- Fully connected layers (MLP)
- Output: Predicted efficacy score (regression)



The Overall Architecture of DeepFM-Crisp

DNA-RNA free energy

Approximate free energy model for gRNA-targetDNA binding¹:

$$\Delta G^B[g,t] = \delta PAM \times (\Delta G^H - \Delta G^O - \Delta G^U),$$

- ΔG^H — gRNA–DNA hybridisation free energy change (calculated using nearest-neighbor RNA–DNA stacking energies with positional Cas9 weights)
- ΔG^O — target DNA–DNA opening free energy change (based on DNA–DNA stacking energies from thermodynamic models)
- ΔG^U — gRNA self-folding free energy change (predicted with RNAfold)

δPAM — correction factor based on PAM sequence (NGG = 1.0, NAG = 0.9, NGA = 0.8)

ΔG^H affects gRNA efficiency²:

- Most efficient gRNAs have ΔG^H values between [−64.53, −47.09] kcal/mol
- Low ΔG^H results in increased activity at off-targets

Code: github.com/rth-tools/crisproff

¹F. Alkan et al. (2018) CRISPR-Cas9 off-targeting assessment with nucleic acid duplex energy parameters [\[Link\]](#)

²G. I. Corsi et al. (2022) CRISPR/Cas9 gRNA activity depends on free energy changes and on the target PAM context [\[Link\]](#)

gRNA-Cas9 docking score

Docking score (DS) quantifies the binding affinity between two molecules (gRNA and Cas protein). A lower (more negative) score indicates a more stable and favorable interaction.

Use of DS for cleavage efficiency prediction:

- Lower DS increases the likelihood of forming an active complex, and therefore **enhances cleavage efficiency**.
- DS does not account for the target DNA — **only the Cas protein and gRNA**.
- It serves as a **preliminary criterion** to identify promising Cas–gRNA pairs from many possible combinations.

CRISPR-Cas-Docker¹ – a web server for optimal crRNA-Cas pair prediction:

AlphaFold + RoseTTAFold → HDOCK² → top-10 models ranked by DS

Code: github.com/hshimlab/CRISPR-Cas-Docker

¹H. Park et al. (2023) CRISPR-Cas-Docker: web-based in silico docking and machine learning-based classification of crRNAs with Cas proteins [[Link](#)]

²Y. Yan et al. (2020) The HDOCK server for integrated protein–protein docking [[Link](#)]

Chromatin accessibility

Chromatin accessibility refers to how open or condensed a genomic region is, affecting the ability of proteins to access DNA.

- Open regions (euchromatin) are typically 100–1000 bp long and associated with active transcription.
- Closed regions (heterochromatin) are compacted and less accessible for DNA-binding proteins.

Impact on CRISPR efficiency¹:

- Cas9 cleavage is more efficient in open chromatin.
- Inaccessible regions often result in reduced editing efficiency due to steric hindrance by nucleosomes.

Chromatin accessibility data

- Public datasets: ENCODE, Roadmap Epigenomics.
- Can be accessed via API

¹K. T. Jensen et al. (2017) Chromatin accessibility and guide sequence secondary structure affect CRISPR-Cas9 gene editing efficiency [[Link](#)]

Code by far

Status	Feature	Based on
Done	Search for potential sgRNAs, efficiency prediction	<u>crispRdesignR</u> (R)
	Off-target event probability prediction	<u>CRISPR-BERT</u> (Python)
	DNA-RNA hybridization energy	<u>CRISPROff</u> (Python)
	Relative activity predictor	<u>ExsgRNA</u> (Python)
	Genome off-target screening	<u>crispRdesignR</u> (R)
In progress	RNA design with reinforcement learning	<i>See the next slide</i>
Other plans	Cas9-gRNA docking score	<u>RF2NA</u> (Python)

Reinforcement Learning for gRNA fine-tuning

Tool	About	What suits us
<u>LEARNA</u> ¹	<i>A PPO-based agent generates RNA one nucleotide at a time so that the final sequence folds into a target structure</i>	<i>Replace the reward with: score = Efficiency(gRNA) - $\lambda \cdot \Delta G_h$; the character-by-character generation and local improvement logic are already suitable</i>
ERLBioSeq ²	<i>A hybrid framework combining an RL generator, evolutionary search, and LGFEN scoring network to design DNA/RNA/proteins without training data</i>	
<u>EdgeAlign</u> ³	<i>A dueling-DQN aligns DNA using match/insert/delete operations in a sliding window and runs efficiently on Jetson/FPGA</i>	
<u>Eastman A3C RNA</u> ⁴	<i>An A3C agent iteratively edits nucleotides until the folded structure matches the target</i>	<i>The "action = point mutation" format fits perfectly for fine-tuning an already decent gRNA based on hybridization energy stability</i>

¹F. Runge et. al. (2019) Learning to design RNA [\[Link\]](#)

²X. Zheng et. al. (2024) Designing Biological Sequences without Prior Knowledge Using Evolutionary Reinforcement Learning [\[Link\]](#)

³A. Lall & S. Tallur (2023) Deep reinforcement learning-based pairwise DNA sequence alignment method compatible with embedded edge devices [\[Link\]](#)

⁴P. Eastman et. al. (2018) Solving the RNA design problem with reinforcement learning [\[Link\]](#)

Next steps

Interim Conclusions

- Hybrid structural + semantic features outperform SOTA with only 2.4 M parameters.
- Workflow already usable for in-silico knock-out screening.
- Off-target and thermodynamic modules plug in seamlessly to ML core.

Next steps

- Train RL fine-tuner on seed/PAM mutagenesis.
- Add docking & chromatin data as live features.
- Release open-source beta with minimal web UI in summer 2025.

THANKS!

I will be glad to answer your questions

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[grna_design_tool](#)



SCAMT

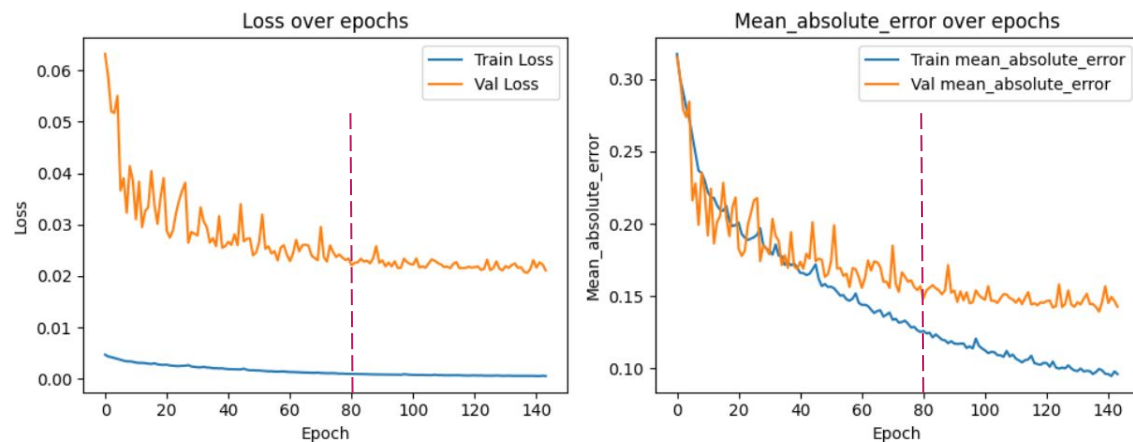
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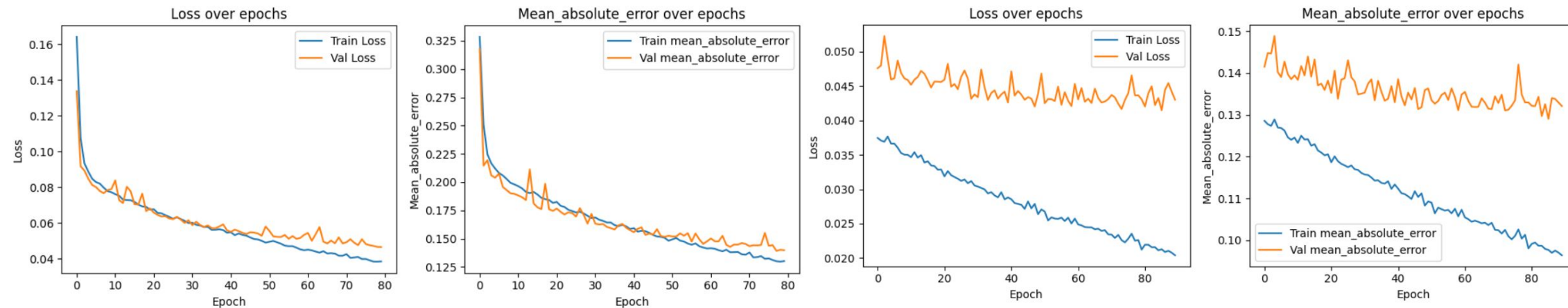
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Overfitting check

ExsgRNA:



Our model:



80 epochs

another 80 epochs. Reached a plateau, stopped training